Physics
Standard and higher level
Markscheme
B.3 Gas laws

12 October 2026

W4ss Academy Paper 2 practice 1 | **Topic** B.3 Gas laws | **Ref.** W4SS–PHY–B3–P2–01

Student name

Paper 2

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- Award one mark per ✓. Accept equivalent answers (OWTTE) and error carried forward (ECF) where stated.
- Total: **[30 marks]**.

Paper 2 — markscheme

Question	Marks	Answers
1. (a)	**[1]**	$p = \frac{nRT}{V}$ seen $= 12$ kPa ✓
1. (b)	**[1]**	6.4×10^{-2} kg ✓
1. (c)	**[1]**	$v = \frac{\sqrt{3PV}}{M} = 450$ m s^{-1} ✓
1. (d)	**[4]**	increased temperature means increased average KE and hence increased average translational speed ✓ This increases the momentum transfer at the walls for each collision / mv is greater per collision ✓ This increases the frequency of collisions at the walls / particles cover the distance between walls more quickly ✓ Ideas that $F_{wall} = \frac{\Delta p}{\Delta t}$ AND $P = \frac{F}{A}$ (can be in words) so that force increases and pressure increases ✓
2. (a)	**[2]**	$1.8 \times 10^5 = \frac{1}{3} \times 2.0 \times v^2$ ✓ $v = 520$ «m s^{-1}» ✓
2. (b)	**[3]**	Average kinetic energy of the molecules is determined by the temperature only ✓ The mass of a molecule is different for each component gas ✓ From $E_k = \frac{1}{2}mv^2$, the same E_k and different mass implies a different average velocity ✓
2. (c)	**[2]**	***ALTERNATIVE 1*** The average translational speed increases «because T increases» ✓ From $P = \frac{1}{3}\rho v^2$, the density decreases «to keep P constant» ✓ ***ALTERNATIVE 2*** From the ideal gas law, the volume of the gas increases ✓ Since $\rho = \frac{m}{V}$ and m is constant, the density decreases ✓
3. (a)	**[1]**	kg m^2 s^{-2} ✓

3. (b)	**[3]**	***ALTERNATIVE 1*** Graph shown is a straight line/linear ***OR*** expected graph should be a straight line/linear ✓ If ideal then T intercept must be at $T = -273$ °C ✓ Use of $y = mx + c$ to show that $x = -273$ °C when $y = 0$ ✓ (hence ideal) ***ALTERNATIVE 2*** Calculates $\frac{pV}{T}$ for two different points ✓ Obtains 1.50 «J K $^{-1}$ » for both ✓ States that for ideal gas $\frac{pV}{T} = nR$ which is constant and concludes that gas is ideal ✓
3. (c)	**[3]**	Use of $n = \frac{pV}{RT}$ ***OR*** $N = \frac{pV}{kT}$ ✓ Mass of gas = $n \times N_A \times$ mass of molecule ***OR*** Mass of gas = N × mass of molecule ✓ 5.1 «g» ✓
4. (a)(i)	**[3]**	energy required = 250 × 4200 × (30 − 15) ✓ energy available = 0.30 × 680 × t × A ✓ $A = \text{«} \frac{250 \times 4200 \times 15}{0.30 \times 680 \times 60 \times 60} =$ » 21 «m ^{2} » ***OR*** 22 «m ^{2} » ✓ *Allow **ECF** from **MP1** and **MP2** .* *Accept the correct use of 0.30 in either **MP1** or **MP2** .*
4. (a)(ii)	**[3]**	absorbed intensity = (1 − 0.2) × 680 «= 544» «W m $^{-2}$ » ***OR*** emitted intensity = 0.97 × 5.67 × 10 $^{-8}$ × T ^{4} ✓ $T = \sqrt[4]{\frac{544}{0.97 \times 5.67 \times 10^{-8}}} = 315$ «K» ✓ 42 «°C» ✓ *Allow **ECF** from **MP1** and **MP2** .* *Allow **MP1** if absorbed or emitted intensity is multiplied by area.*
4. (b)(i)	**[1]**	can be liquefied ✓ has intermolecular forces / potential energy ✓ has atoms/molecules that are not point objects / take up volume ✓ does not follow the ideal gas law «for all T and p » ✓ collisions between particles are non-elastic ✓ *Accept the converse argument.*

4. (b)(ii)	**[2]**	***ALTERNATIVE 1*** «constant *p* and *V* imply» *nT* = const ✓ *T* increases hence *n* decreases ✓ ***ALTERNATIVE 2*** «constant *p* and *n* imply» *V* is proportional to *T* / air expands as it is heated ✓ «original» air occupies a greater volume **OR** some air leaves through opening ✓ ***MP2** in **ALT 2** must come from expansion of air, not from expansion of water.* *Award **[0]** for an answer based on expansion of water.* *Award **[1] max** for an answer based on convection currents.*